# PERFORMANCE CHARECTERISTICS OF SELF PRIMING PUMP.

**AIM:-**

To study the performance characteristics of a self-priming pump and plot the following graphs.

1. Discharge v/s Input power
2. Discharge v/s efficiency
3. Dischaige v/s Head

**APPARATUS USED**

1. Self-priming pump with suction and delivery pipes and necessary valves.
2. A measuring tank fitted with a piezometer tube and drain valve.
3. An energy meter fitted in the apparatus.
4. A stop watch.

**PRINCIPLE**

Efficiency of the pump is the ratio between the power output from the pump and the input to the pump. The power output from the pump is directly proportional to the total head and discharge. The power input of A.C motor is measured with the help of the energy meter and the overall efficiency is calculated.

Efficiency of the pump, $\eta$ $= \frac{Output}{Input} \times 100\ \%$

Output power, $OP$ $= \frac{\rho\, g\, Q\, H}{1000}$ kW

Where

$\rho$ = density of water in $kg/m^3$

g = acceleration due to gravity in $m/s^2$

$Q$ = discharge of water in $m^3/s$

$H$ = total head in m

Discharge, $Q$ $= \frac{A\, R}{t}$ $m^3/s$

Where

A = area of collecting tank in $m^2$ (0.698 x 0.448)

R = rise of water level in the collecting tank in m.

## OBSERVATIONS

| | Delivery pressure | | Suction pressure | | Total Head H | Time for 10 Impulses. 'T' | Time for 5 cm t | Actual Discharge Q | Output power | Input power | Efficiency |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | $kg/cm^2$ | m of water | mm of Hg | m of water | m of water | sec | sec | $m^3/s$ | kW | kW | % |
| 1 | | | | | | | | | | | |
| 2 | | | | | | | | | | | |
| 3 | | | | | | | | | | | |
| 4 | | | | | | | | | | | |
| 5 | | | | | | | | | | | |

| | | |
|---|---|---|
| t | = | time in seconds for rise of liquid level 'R' |

Total head, $H$ = Suction head + Delivery head + Datum head in m of water

Datum head = 0.15 m (level difference of tapping points of pressure gauges)

Input power, $IP$

$$= \frac{n\,3600}{T\,E_m}\,\eta_m \text{ kW}$$

Where

$n$ = Number of revolution of energy meter.
$T$ = Time for 'n' revolution of energy meter in sec.
$E_m$ = Energy meter constant in rev/kW hour.
$\eta_m$ = Efficiency of the motor.

## PRECAUTIONS

1. Never run the pump without liquid in it as this would cause damage to stuffing box, bush bearing etc.
2. Never try to throttle the suction side of the pump to control discharge as it would seriously affect the performance of the pump.

## PROCEDURE

1. Close the delivery valve fully and start the motor. The valves of gauges to be closed during starting.
2. Open the delivery valve fully and allow the flow to stabilize, open the pressure gauge and vacuum gauge valves.
3. Adjust the delivery valve to bring the pressure to 5 kg/cm$^2$. Note the pressure gauge and vacuum gauge readings.
4. Note the time 't' for 5 cm rise of liquid in the measuring tank using a stop watch.
5. Also note the time 'T' for 10 impulses of the energy meter to calculate the input power.
6. The above procedure is repeated by gradually decreasing the pressure (say 4, 3, 2, 1 kg/cm$^2$).
7. Switch off the power supply.

## RESULT

1. The overall efficiency of the ~~gear~~ self priming pump is ........%.

## INFERENCE

## SAMPLE CALCULATION (Set no:.......)

Input power, I.P $= \frac{n\ 3600}{T\ Em}\eta_m$ kW

Number of impulses of energy meter, n = 10 Impulse

Time for 'n' rev of the energy meter, T = ...l......... s

Energy meter constant, $E_m$ = 3200 Imp/kW hour

Mechanical efficiency, $\eta_m$ = 80 %

Input power, IP = kW

Output power, OP $= \frac{\rho\ g\ H_{total}\ Q}{1000}$ kW

Acceleration due to gravity, g $= 9.81\ \frac{m}{s^2}$

Total Head = Suction Pressure + Delivery Pressure + Datum Head in m of water

Delivery Pressure head $= \text{.........}\ \frac{kg}{cm^2} = \text{......} \times 10 = \text{........}$ m of water

Suction Pressure head $= \text{........}$ mm of Hg $= \frac{\text{.......}}{1000} \times 13.6 = \text{........}$ m of water

Datum Head $= 150\text{mm} = \frac{150}{1000} = 0.150$ m of water

Total Head = ......................m of water

Discharge, Q $= \frac{A\ H}{t}\ \frac{m^3}{s}$

Area of collecting tank, A $= 0.698 \times 0.448\ m^2$

Height of water collected, H = 0.05 m

Time for 5cm of rise of water in the collecting tank, t= ...............sec

Discharge, Q $= \frac{AH}{t}\ \frac{m^3}{s} = \frac{0.698 \times 0.448 \times 0.05}{'t'} = \ \frac{m^3}{s}$

Output power = ............................... kW

Overall Efficiency of Self priming pump $= \frac{Output}{Input} \times 100\ \%$